Quantum Zeno and Anti-Zeno Physics: Suppressing and Enhancing Quantum State Evolution

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We review the theory of the Quantum Zeno and Anti-Zeno effects, the suppression (Zeno) or enhancement (Anti-Zeno) of transition rates induced by measurement backaction. We discuss both the simple (or "naive") picture for realizing the Zeno effect as well as a more robust formalism that readily explains both the Zeno and Anti-Zeno effects. We then discuss two experimental studies that illustrate physical realizations of these effects. We expand upon these experimental works by analyzing their results using the aforementioned Zeno Physics theoretical frameworks. Finally, we introduce the Stochastic Master Equation and Monte-Carlo Wavefunction method as tools to study individual quantum trajectories, and their uses for studying Zeno effects. We supplement this discussion with numerical simulations of the Stochastic Master Equation using the toolkit QuTiP and use these simulations to observe the Quantum Zeno effect.

I. INTRODUCTION

The Quantum Zeno Effect (QZE) is a phenomenon in which frequent measurements suppress the evolution of a quantum state, effectively freezing it and inhibiting transitions to other accessible states by repeatedly projecting it back onto itself. While the QZE can be intuitively understood as repeated wavefunction collapse into a specific state, this picture is incomplete. In fact, under certain conditions and arguably more readily observed, frequent measurements can instead expedite the evolution of a quantum state, accelerating the system's decay. This phenomenon is known as the Quantum Anti-Zeno Effect (QAZE). In this work, we introduce the frameworks used to understand the Q(A)ZE and review experimental realizations of these effects. Furthermore, we will discuss methodologies to numerically study the Zeno-Effect, and more generally measurement-modified quantum dynamics, via the Stochastic Master Equation and the Monte-Carlo Wavefunction Method.

II. THEORY OF THE QUANTUM ZENO AND ANTI-ZENO EFFECTS

A. The "Naive" Picture

The Zeno effect can be elegantly derived from the most basic of quantum mechanics. It will turn out that this "naive" approach provides an incomplete picture of the Zeno effect, however, it is nonetheless quite instructive. Consider a quantum state, which at t=0, is given by $|\Psi(0)\rangle = |e\rangle$. Of course, the time-evolved state is then obtained by acting on our state with the time-evolution operator, $|\Psi(t)\rangle = \hat{U}|e\rangle = e^{-i\hat{H}t/\hbar}|e\rangle$. The probability to measure our state at a later time t is given by Equation 1, where the approximate form on the right is found by expanding around small times [1]:

$$P_{ee}(t) = |\langle e|e^{-i\hat{H}t/\hbar}|e\rangle|^2 \approx 1 - \langle V^2\rangle t^2/\hbar^2 \qquad (1)$$

where $\langle V^2\rangle = \langle H^2\rangle - \langle H\rangle^2$, the variance of the Hamiltonian operator. Now we consider (ideally) measuring our system n times in regularly-spaced intervals of $t = \tau$, or in other words, measuring our system repeatedly with measurement rate $\nu \sim 1/\tau$, The probability $P_{ee}(t = n\tau)$ is given by Equation 2, where the approximate form on the right is found in the limit of $\sqrt{\langle V^2\rangle} \ll \nu$ [1].

$$P_{ee}(t = n\tau) = (1 - \langle V^2\rangle \frac{t}{\nu\hbar^2})^n \approx e^{-\hbar^{-2}\langle V^2\rangle \frac{t}{\nu}} \qquad (2)$$

Figure 1 plots $P_{ee}(t = n\tau)$ for a various ratios of $\sqrt{\langle V^2\rangle}/\nu$. The result illustrates that with sufficiently high measurement rates, we can effectively "freeze" our system into the $|e\rangle$ state. More formally, with a high enough measurement rate, we repeatedly project our quantum state back into the measured basis state before it has time to coherently evolve into other states – manifesting as a longer lifetime of $|e\rangle$. This is precisely the Quantum Zeno effect.

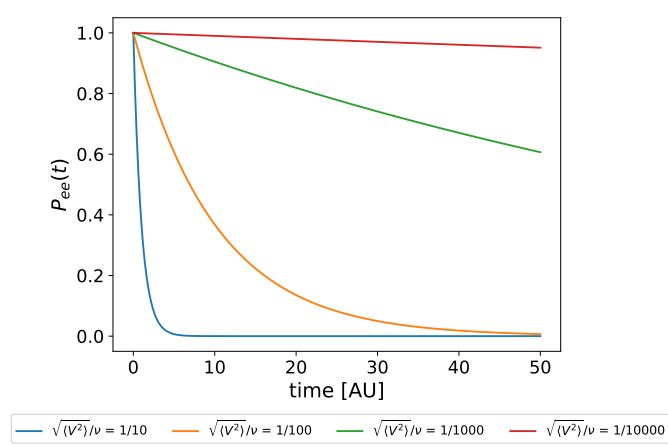

FIG. 1. $P_{ee}(t)$ for various ratios of $\sqrt{\langle V^2\rangle}/\nu$. As the measurement rate becomes sufficiently larger than $\sqrt{\langle V^2\rangle}$, the $|e\rangle$ lifetime increases.

B. Zeno and Anti-Zeno effects via the "Overlap Integral" Formalism

As mentioned before, and implied by us calling the prior derivation of the Zeno effect the "Naive" picture, there is an underlying problem with the above approach – it assumes idealized measurements. In reality, our measurements are subject to the Energy-Time Uncertainty Principle, $\Delta E \Delta t \sim \hbar$. If we take $\Delta t \sim 1/\nu$, one can easily see that increasing our measurement rate is not without consequence. As we increase the measurement rate, the transition energy of our state spreads accordingly, which can lead to our state coupling to undesired decay channels. To address this we turn to the formalism first derived by Kofman and Kurizki [1], which we will refer to as the overlap integral formalism in this work. We begin by considering the Hamiltonian $\hat{H} = \hat{H}_0 + \hbar \sum_j (V_{ej}|e\rangle\langle j| + V_{je}|j\rangle\langle e|)$. Where the second term accounts for interactions between $|e\rangle$ and the $\{|j\rangle\}$ states. Taking both $|e\rangle$ and the $\{|j\rangle\}$ to be eigenstates of $\hat{H}_0$, we can write the most general form of our wavefunction to be $|\Psi(t)\rangle = \alpha(t)e^{-i\omega_a t}|e\rangle + \sum_j \beta_j(t)e^{-i\omega_j t}|j\rangle$ [1]. Solving the Schrödinger Equation in the short-time behavior limit, $\alpha(t) \approx \alpha(0) = 1$ and finding $P_{ee}(t = n\tau)$ yields Equation 3, where R, in the frequency-domain, is given by Equation 4 [1].

$$P_{ee}(t = n\tau) = |\alpha(t)|^{2n} \approx e^{-Rt} \tag{3}$$

$$R = 2\pi \int_0^\infty d\omega G(\omega)F(\omega) \tag{4}$$

Where $G(\omega)$ is the spectral density of the $|j\rangle$ (some reservoir our system is coupled to) states weighted by their coupling strengths to $|e\rangle$, $|V_{ej}|^2/\hbar^2$. $F(\omega)$ is the broadened transition spectrum of $|e\rangle$ with linewidth ν accounting for effects due to Energy-Time Uncertainty Principle from frequent measurements [1]. R is simply the effective decay rate of $|e\rangle$ (lifetime $\tau = 1/R$). To expand upon the arguments of Kofman and Kurizki more explicitly, very infrequent measurements would then correspond approximately to Equation 5, which can simply be taken as the "bare" decay rate.

$$R_0 = 2\pi \int_0^\infty d\omega \delta(\omega - \omega_0)G(\omega) = 2\pi G(\omega_0) \tag{5}$$

What can this tell us about when the Zeno effect emerges? If $R < R_0$, then the decay rate has decreased from its "bare" value and hence we observe the Zeno effect. However, we have also found another regime, $R > R_0$, which corresponds the decay rate increasing (shorter lifetime). We have found the phenomena known as the Quantum Anti-Zeno effect (QAZE)! The authors make the graphical arguments that the QZE is expected if $\nu \gg \Gamma_R, |\omega_a - \omega_m|$ where ν is the measurement rate (and hence the linewidth of $F(\omega)$), Γ_R is the width of $G(\omega)$, ω_a is the transition frequency of $|e\rangle$, and ω_m is the peak frequency of the weighted bath spectrum [1]. The QAZE is then expected in the limit of $\nu \ll |\omega_a - \omega_m|$. We believe these arguments, while instructive for building intuition, are too much of a generalization, and a simple comparison between R and R_0 is better practice. We will do this explicitly to analyze experimental results later in this work.

III. EXPERIMENTAL REALIZATIONS OF Q(A)ZE PHYSICS

A. How to Stop Atoms from Tunneling

An interesting realization of the QZE was illustrated by Patil et al. (2015) [2]. In this experiment, the authors study ultracold gas confined to an optical lattice. The atoms of the gas coherently tunnel across lattice sites with tunneling rate J. Such tunneling can lead to multiple atoms occupying the same lattice site, which can then lead to atom loss via photo-association with rate κ [2]. The authors observe the position of the atoms with continuous position measurements (or imaging) by scattering photons off the atoms with a measurement rate, Γ_m [3]. The authors find two main regimes. The first is given by limit $\Gamma_m \ll J$, in which case the authors find that both the tunneling rate and the photo-association are not affected by the measurements. The second regime is $\Gamma_m \gg J$. In this limit, the authors find that the tunneling rate decreases and takes an effective form of $\tilde{J}_{\text{eff}} \sim J^2/\Gamma_m$. Consequentially, the photo-association rate decreases as well.

In this work, we now expand upon the work of Patil et al. (2015) by analyzing the authors' results using the "naive" QZE picture as described above. Let's consider a two lattice-site system where an atom in the left lattice-site is given by state $|e\rangle$ and an atom in the right lattice-site is given by state $|f\rangle$. Considering only the tunneling dynamics, the Hamiltonian of such a system can be given by $\hat{H} \sim -J(\hat{a}^\dagger\hat{b} + \hat{b}^\dagger\hat{a})$ (taking $\hbar = 1$). Where $\hat{a}^\dagger$ is the creation operator of an atom in the left lattice and $\hat{b}^\dagger$ in the right lattice. J is the tunneling rate, using the same notation as Patil et al. (2015). The variance of the Hamiltonian with respect to state $|e\rangle$ is easily calculated to be $\langle V^2\rangle \sim J^2$. Now plugging this into Equation 2, using $\nu \equiv \Gamma_m$, we find Equation 6.

$$P_{ee}(t = n\tau) \approx e^{-J^2 t/\Gamma_m} \tag{6}$$

From our above discussion of the naive picture, we expect to see the QZE in the limit of $\sqrt{\langle V^2\rangle} \ll \nu \to J \ll \Gamma_m$. Beautifully, we have derived the exact same dynamics found experimentally by Patil et al. (2015) including the effective tunneling (decay) rate of an atom in a given

lattice site, $R = \tilde{J}_{\text{eff}} \sim J^2/\Gamma_m$. This brief derivation illustrates that while the naive picture has problems, it can still be used to describe real phenomena for the correct physical setup.

B. How to Make a Good Qubit Go Bad

Another interesting realization of both the Zeno and Anti-Zeno effect is that of Harrington et al. (2017). The authors of this paper study the Q(A)ZE in a superconducting qubit system by introducing a synthesized Lorentzian amplitude spectrum from a white noise source to mimic some thermal bath that the qubit couples to [4]. The bath can be controllably detuned from the qubit's transition frequency (ω_{ge}, hence controlling which modes of the bath the qubit can readily couple to. The authors measure the qubit using a typical circuit QED quantum non-demolition measurement via dispersive readout of the resonator coupled to the qubit. Measuring the qubit through the resonator in this fashion allows one to gather information about the qubit's state in the energy eigenstate basis [4]. The authors sweep the detuning between the artificial bath and the qubit and measure the system at two different rates, 1 MHz and 2 MHz, and measure T_1 (the qubit's excited state lifetime). The results show that whether the Zeno (increased T_1) or Anti-Zeno (decreased T_1) effect is observed depends on the detuning between the bath and qubit as illustrated in Figure 2 (Figure 2(C) from [4]).

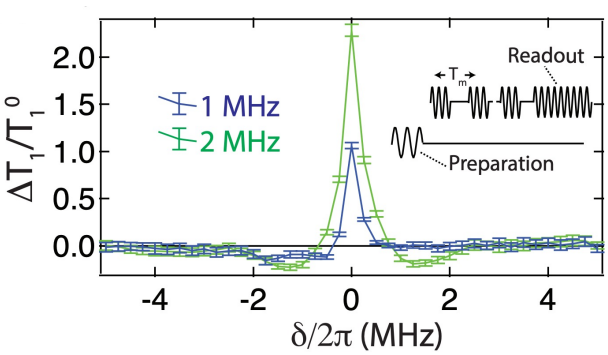

FIG. 2. Change in the $T_1(\frac{\Delta T_1}{T_1^0})$ of the qubit as a function of detuning from the bath. This is taken from Figure 2(C) from [4].

The vertical-axis of Figure 2 is $\frac{\Delta T_1}{T_1^0} = \frac{T_1 - T_1^0}{T_1^0}$, where T_1 is the effective lifetime and T_1^0 is the lifetime in absence of any Q(A)ZE. Thus the Zeno effect is observed when $\frac{\Delta T_1}{T_1^0} > 0$ and the Anti-Zeno when $\frac{\Delta T_1}{T_1^0} < 0$. We see that when the bath and qubit are very close in resonance $|\delta|1$ MHz, the Zeno-Effect is observed. Past this detuning limit, the Anti-Zeno effect is observed until about $|\delta|4$ MHz, where the measurement rate no longer perturbs the "bare" evolution of the system (e.g. $\frac{\Delta T_1}{T_1^0} \approx 0$). The authors explain this behavior via the overlap formalism introduced above: as $G(\omega)$ (the weighted bath spectrum) varies in frequency, the broadened qubit transition spectrum's, $F(\omega)$, overlap with $G(\omega)$ increases or decreases and hence the effective qubit decay rate, $R = 1/T_1$, increases or decreases accordingly.

In this work, we will now expand upon this explanation of Harrington et al. (2017) by numerically calculating the overlap integrals. Since we do not have access to the direct data from this paper, we reconstruct the bath spectrum as a Gaussian that approximately has the same full-width half-max as the bath spectrum shown in the paper. We then compare the overlap of the bath spectrum with an idealized narrow transition spectrum (approximated as a delta function) and with broadened spectra $F(\omega)$ corresponding to $\Gamma_m = 1, 2$ MHz. A visual example of this sweep for a few detuning points is shown in Figure 3. We then plot the difference in the overlap integrals, $R_0 - R$ (bare decay rate minus the Zeno-modified decay rate) as a function of detuning, Figure 4. From these numerical calculations of the overlap integrals, we find that $|\delta|1$ MHz corresponds to $(R_0 - R) > 0$, meaning the decay rate of the qubit has decreased (T_1 increased) and hence this region corresponds to the Zeno-Effect. We observe $(R_0 - R) < 0$ (T_1 decreased), the Anti-Zeno effect, until about $|\delta|4$ MHz where $(R_0 - R)$ finally goes to 0 and we observe no Zeno or Anti-Zeno physics. Our numerical calculations are in very strong agreement with the experimental results of Harrington et al. (2017).

Finally, the authors study whether the Q(A)ZE is dependent on a measurement projecting our state onto an energy eigenstate (i.e. does it depend on wavefunction collapse physics?). To test this the authors perform "quasi"-measurements of the system by purely dephasing the qubit without actually accumulating any information about the energy states. In interest of space, we will not discuss this result in depth but the punchline is that the authors find that this pure dephasing "quasi"-measurements yield almost identical Zeno and anti-Zeno behavior as shown in Figure 2 from true dispersive measurements. This demonstrates that state projection is not required to observe the Q(A)ZE, instead, the effect arises purely from interaction-induced spectral broadening of the qubit and its resulting overlap with the bath.

IV. STUDYING INDIVIDUAL QUANTUM TRAJECTORIES

Zeno and Anti-Zeno effects are fundamentally measurement/interaction phenomena. We need formalism(s) that describe quantum state evolution conditioned on measurement outcomes to properly study these effects. Two prominent such methods are stochastic master equations (SME) and the Monte-Carlo Wavefunction Method (MCWF).

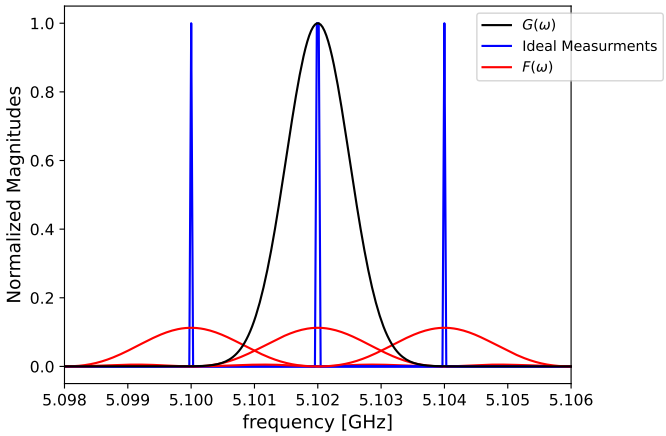

FIG. 3. An example of ideal measurements and $F(\omega)$, with $\Gamma_m = 1$ MHz, at overlap points (or detuning) with the weighted bath spectrum. The magnitudes of these functions are normalized.

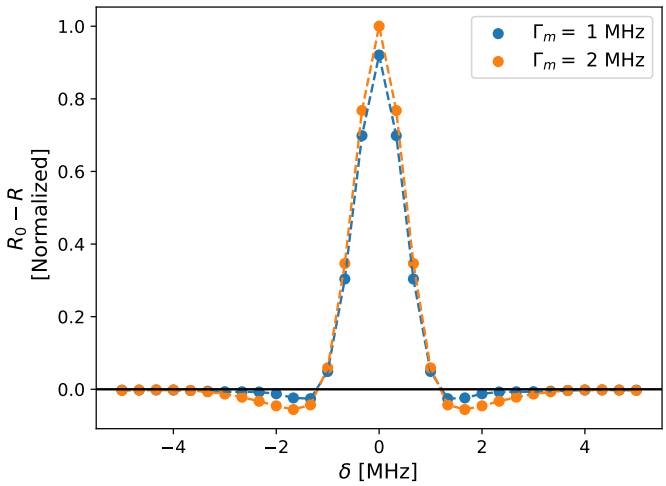

FIG. 4. Change in the decay rate, $R_0 - R$, calculated from overlap integrals (Equations 4 & 5) with the weighted bath spectrum.

A. Stochastic Master Equations

Open-quantum systems are subject to various decay and decoherence mechanisms due to interactions with their environment. In practice, one can only continuously monitor some of these decay channels. For instance, in a circuit QED architecture, such as a coupled resonator-qubit system, there are four decay channels of consideration. There is the resonator photon loss rates, κ_{in} and κ_{out}, which describe the rate of photon loss from the input and output ports of the resonator [5]. For the qubit, there is the qubit decay rate (γ_1) and dephasing rate (γ_ϕ) [5]. Researchers can typically only measure κ_{out} – hence for such a system, the dynamics are conditioned upon the continuous measurements of photons leaving the resonator output port as done by Gambetta et al. (2008) assuming homodyne detection [5]. We can study these conditioned dynamics using an SME. In the SME formalism rather than considering $\rho(t)$, the solution to the Master Equation, one considers the dynamics of $\rho_J(t)$, the system density matrix conditioned on continuous measurements, $J(t)$. This, of course, is where the stochasticism of the "stochastic master equation" comes from, since quantum measurement outcomes are inherently random. Furthermore, it is easy to see how such an equation would be useful for studying the Zeno effect, but we will touch base more on this shortly.

Each solution of the SME for a given random measurement record, $J(t)$, defines a stochastic quantum trajectory, the solution to the typical master equation is the ensemble average of all these possible quantum trajectories, $\rho(t) = E[\rho_J(t)]$. An important note is that if one could measure all decay channels precisely then the system conditioned on $J(t)$ would simply be a pure state, $|\Psi_J(t)\rangle$ [5]. Now continuing with Gambetta et al. (2008), the authors are dealing with a resonator-qubit system but the the actual system of interest is the qubit alone. The authors begin by deriving the total SME for the resonator-qubit system and then use a polaron-like transformation to obtain a qubit-only SME. The resulting qubit-only SME is given in Equation 7 [5].

$$\begin{aligned}\dot{\rho}_{\bar{J}}(t) = \mathcal{L}\rho_{\bar{J}}(t) \\ &+ \sqrt{\Gamma_{ci}(t)}\,\mathcal{M}[\sigma_z]\rho_{\bar{J}}(t)\left(\bar{J}(t) - \sqrt{\Gamma_{ci}(t)}\langle\sigma_z\rangle_t\right) \\ &- i\,\frac{\sqrt{\Gamma_{ba}(t)}}{2}[\sigma_z, \rho_{\bar{J}}(t)]\left(\bar{J}(t) - \sqrt{\Gamma_{ci}(t)}\langle\sigma_z\rangle_t\right).\end{aligned} \tag{7}$$

Let's breakdown each term here to understand how it contributes to the evolution of the system. The first term $\mathcal{L}\rho_{\bar{J}}(t)$ is the Lindbladian (i.e. normal master equation evolution). The measurement term is $\sqrt{\Gamma_{ci}(t)}\,\mathcal{M}[\sigma_z]\rho_{\bar{J}}(t)(\bar{J}(t) - \sqrt{\Gamma_{ci}(t)}\langle\sigma_z\rangle_t)$. The homodyne gain term, $\sqrt{\Gamma_{ci}(t)}\,\mathcal{M}[\sigma_z]\rho_{\bar{J}}(t)$, collapses the state towards eigenstates of σ_z. Where $\mathcal{M}[c]$ is the measurement superoperator and Γ_{ci} is the rate at which coherent information comes out of the resonator [5]. The last term in Equation 7 is the noisy, non-Heisenberg backaction term which is induced by measurements, but does not add any information gain [5]. This term corresponds to random phase kicks around the z-axis on the Bloch Sphere. Γ_{ba} is the measurement backaction rate. Numerically solving this equation can allow us to vary these two measurement-related rates and see how Zeno-Effects arise.

Using QuTiP, we solve the SME for individual quantum trajectories and extract $P_{ee}(t)$ [6]. We do this for a qubit with no drive in two different regimes, $\Gamma_m \ll \gamma_1$ and $\Gamma_m \gg \gamma_1$ (Figure 5). We also do this for a qubit driven with a drive $\hat{V} = \frac{1}{2}\Omega_r\sigma_x$ in the regimes, $\Gamma_m \ll \gamma_1, \Omega_r$ and $\Gamma_m \gg \gamma_1, \Omega_r$ (Figure 6). QuTiP does not solve exactly

the same SME as that shown in [5]; rather, it solves a related stochastic equation that does not explicitly include the additional phase-backaction term written in Eq. 7.

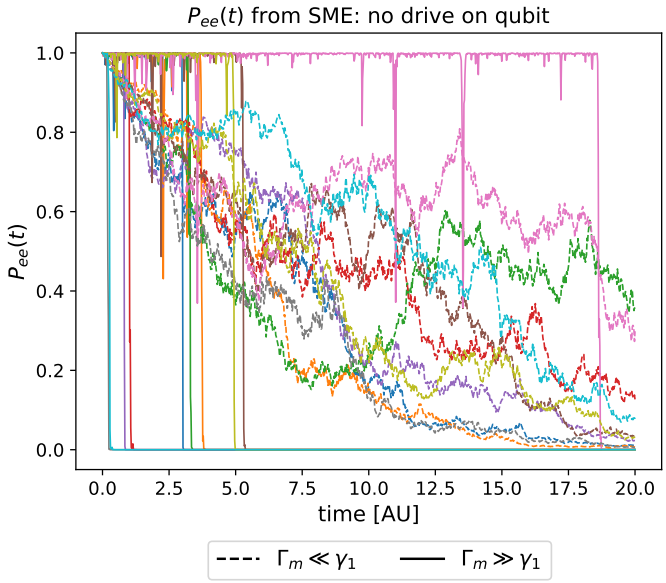

FIG. 5. $P_{ee}(t)$ of ten individual quantum trajectories via numerically solving the SME with QuTiP for no drive on the qubit. The initial condition is $\rho_J(0) = |e\rangle\langle e|$. The dashed-lines correspond to solving the SME in the regime of $\Gamma_m \ll \gamma_1$ and the solid-lines to $\Gamma_m \gg \gamma_1$

.

As shown in Figure 5, when the decay rate is much larger than measurement rate, the quantum trajectories of our system show expected T_1 relaxation. On the other hand, if the measurement rate is much larger than the decay rate, we see "freezing" of our qubit in the first excited state, corresponding to repeated projection of our system into the $|e\rangle$ state. This is the Quantum Zeno effect. In Figure 6, we observe similar behavior. When the measurement rate is much weaker than both the relaxation rate and qubit drive strength, the quantum trajectories show typical, noisy Rabi Oscillations. On the other, we observe Zeno effects when the measurement rate is much stronger than both the relaxation time and qubit drive strength. In both of these numerical simulations, notice we observe some "quantum jumps" that occur when a measurement happens to collapse our state into the ground state. We speculate that one could likely observe the Anti-Zeno effect as well with this framework if the full-set of decay channels was integrated into the SME. In the examples above we only integrated γ_1 decay. We also neglected dephasing in the above example for simplicity, although we plan to integrate dephasing and other decoherence terms in the future (out of my own pure interest since this paper has a deadline).

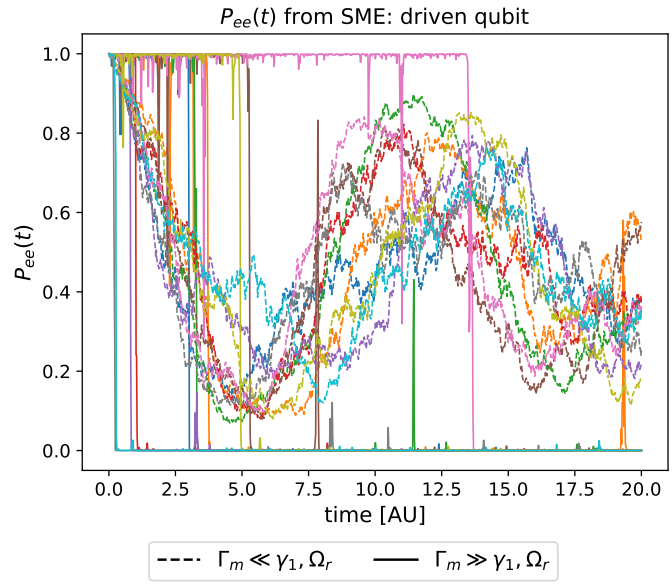

FIG. 6. $P_{ee}(t)$ of ten individual quantum trajectories via numerically solving the SME with QuTiP for a driven qubit. The initial condition is $\rho_J(0) = |e\rangle\langle e|$. The dashed-lines correspond to solving the SME in the regime of $\Gamma_m \ll \gamma_1, \Omega_r$ and the solid-lines to $\Gamma_m \gg \gamma_1, \Omega_r$

. In both cases, we consider a strong drive, $\Omega_r > \gamma_1$.

B. Monte-Carlo Wavefunction Method

Finally, we briefly introduce another method used to study open quantum systems, which is particularly relevant for the Zeno effect. The Monte-Carlo Wavefunction (MCWF) method, or Quantum Jump Method, stochastically evolves a single pure state coupled to a reservoir [7]. The basis of this method begins by considering a non-Hermitian Hamiltonian $\hat{H} = \hat{H}_s - i\hbar/2\sum_m \hat{C}_m^\dagger \hat{C}_m$ [7]. Where $\hat{H}_s$ is our normal system Hamiltonian and $\hat{C}_m$ are jump operators, which encode a variety of interactions between our system and the reservoir or measurement-induced wavefunction collapse. Circling back to our above discussions on the Zeno-Effect, one possible measurement-related jump operator is $C_1 = \sqrt{\Gamma_m}\,|e\rangle\langle e|$, which models repeated monitoring of whether the system occupies the excited state. A pure state is numerically evolved using the corresponding time-evolution operator of this effective Hamiltonian in discrete time-steps, Equation 8.

$$|\phi^{(1)}(t+dt)\rangle = (1 - \frac{i}{\hbar}\hat{H}dt)|\phi(t)\rangle \tag{8}$$

Since we are evolving our state under a non-Hermitian Hamiltonian, our pure state is, of course, not normalized. The norm of the dt evolved state is $\langle\phi^{(1)}(t+dt)|\phi^{(1)}(t+dt)\rangle = 1 - dp$, where $dp = \sum_m dp_m = \sum_m dt\langle\phi(t)|\hat{C}_m^\dagger\hat{C}_m|\phi(t)\rangle$ [7]. Now one must check if our state underwent a quantum jump, some interaction with

the environment or information gain from a measurement. To do so, we generate a random number $\epsilon \in [0, 1]$. If $\epsilon > dp$, no jump occurs our system evolves smoothly in this time-step and our normalized wavefunction is $|\phi(t + dt)\rangle = |\phi^{(1)}(t + dt)\rangle/\sqrt{1 - dp}$.

If $\epsilon < dp$, our system underwent a quantum jump. However, we know there are multiple channels from which this can happen encoded by all the different jump operators. We see which jump operator acts on our state during this time-step by randomly picking it via the probability law $\Pi_m = dp_m/dp$. Our evolved state is then given by $|\phi(t + dt)\rangle = \hat{C}_m|\phi(t)\rangle/\sqrt{dp_m/dt}$. Now, say we projectively measure our state at a very high rate, $\Gamma_m \gg 1$, then we expect $dp_1 = dt\sqrt{\Gamma_m}\langle\phi(t)|e\rangle\langle e|\phi(t)\rangle$ to dominate this probability law and hence, we would expect the Zeno effect to emerge via quantum jumps projecting our system back into $|e\rangle$. Or, based on the results briefly discussed in [4], we would also expect Zeno effects to emerge from a dephasing mechanism with a sufficiently large rate as well.

V. ARE QUANTUM TRAJECTORIES REAL?

I was posed the question "Are Quantum Trajectories Real?" during my presentation of this work. After pondering this question a bit more, I believe quantum trajectories are indeed real, but I will dress this remark with a caveat. First, why do I think they are real? A single quantum trajectory represents a specific *possible* evolution of a state due to all the interactions with a coupled bath or by a human measuring it. This really happens in a laboratory, not just in Hilbert space. However, knowing the exact trajectory a real quantum state goes through is, of course, impossible. In this sense, the individual quantum trajectories produced in our SME simulations are not themselves the realized physical trajectory, but rather one of many possible evolutions consistent with the underlying physics.

VI. CONCLUSION

We reviewed the mathematical and physical frameworks that describe the Quantum Zeno and Anti-Zeno effects, including a simplified framework we dubbed the "naive" picture and a more robust formalism, the Overlap Integral Formalism, originally developed by Kofman and Kurizki. We then used both frameworks to analyze two experimental studies of the Q(A)ZE, finding strong agreement with the original results. Finally, we discussed the Stochastic Master Equation (SME) and Monte-Carlo Wavefunction (MCWF) methods as useful approaches for studying Zeno physics, and more broadly quantum trajectories conditioned on measurement backaction.

CODE AVAILABILITY

Numerical calculation and QuTiP simulation codes can be found here PHY-450 Zeno Physics Repository

THANKS FOR A GREAT QUARTER!

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